

Energy Econ Notes 2026

Prof Jacobson

This is as an informal discussion of the characteristics and the pros and cons of the various energy sources of greatest interest to us, from the perspective of what we know today and today's concerns. In this document I write in general about energy and provide my perspective (which, as always, should be taken with a grain of salt). For more specific data and trends, see reports by and resources from bodies like the Energy Information Administration of the United States, the International Energy Agency, and Resources for the Future (many of which I point to in the reading list). By the same token, don't cite this reading as one of your sources in a paper you write – cite something that has been published / reviewed, like the above-mentioned.

Energy Choices

Economists generally believe that people and firms buy the cheapest inputs or items for any given use we can find. This applies to energy just as to everything else. The implication of this is that the pattern of energy use we observe should allow us to infer the relative (market) prices of different kinds of energy. (Remember that externalities are not reflected in these market prices, which gives rise to most the problems we are concerned with in energy.) In other words, the resources we use a lot likely provide cheap energy; the resources we don't use much must be more expensive for the same amount of energy. So if you ask what kind of energy we use, the answer is generally, "cheap energy," and when we use different energy sources for different purposes that's because different things are more and less costly in different locations or contexts.

Because we think people are so driven to increase profits and/or decrease costs, it seems unlikely to me that miraculous low-cost alternative energy sources are hidden in plain view.¹ If someone tells you there is an overlooked energy source that would solve all of our problems, ask yourself why it hasn't gotten much attention yet. Such a source would be very profitable, and while no-one is perfectly rational, I can't imagine firms leaving that much money on the table. (Indeed, big oil companies have often diversified their portfolios to include energy more broadly.) If an energy source is overlooked, it's probably overlooked for a reason.

Different energy sources are suited to different uses. One way to divide up the uses is into sectors such as transportation, industrial, commercial, and residential. Sometimes electricity is pulled out as a separate sector, while sometimes it is folded into categories like those mentioned. These sectors have features that are relevant to thinking about transitions. Transportation has long relied on liquid fuels because batteries are heavy; the transition to greener transportation will require either green liquid fuels or, what seems more likely now for cars and trucks, electrification with lighter batteries. (Aircraft are a problem; battery weight is a huge problem when you're trying to fly, and sustainable aviation fuel doesn't seem super convincing to me but I know people are working on it.) Industrial uses (which include mining,

¹ Could it be that corporate interests (like oil) exert influence to retain their dominance? Personally, I am increasingly convinced by this idea. However, it seems likely to me that that mostly happens through influencing policy which effectively reduces their costs of operating, e.g., by yielding subsidies or by preventing climate regulation. Either way, this still makes oil win by being cheap.

manufacturing, agriculture, and construction) are tricky because concrete and other building materials are very energy intensive and concrete also generates emissions as part of chemical processes separate from energy use. This is why we need to think about embodied energy in products and buildings.

Electricity will become increasingly important because electrification is our main pathway to greening things like transportation and heating. The mix of electricity sources varies quite a bit from country to country and across recent time. Coal used to be the most important electricity fuel; natural gas became increasingly important in recent decades; renewables have been growing tremendously and we can hope they will grow even more; and nuclear has been largely holding steady over recent decades. You can check the current mix of energy on the grid that serves the Berkshires by looking at the ISO New England website.

What kind of energy will we use in the future? Right now, quantitatively, fossil fuels are still the world's largest sources of energy, and they are nonrenewable in the sense that they don't regenerate at a meaningful rate. However, they are all still very plentiful, and we are continuing to explore and find more reserves. It is true that we may have found a lot of the easiest-to-exploit mines and wells already (for some resources, we may have picked a lot of the "low-hanging fruit"). At some point these fuels will show increasing scarcity, and even now we're finding ourselves exploiting increasingly hard-to-get-at resources. Thus, marginal extraction costs will probably continue to rise even as extraction technology improves. What will happen as a result?

First, the prices of energy sources, and the price of "energy services" (basically, the price users face to use energy) will increase. Energy prices are always volatile so may go up and down at any given moment, but the trend should be upward.

This increasing price will cause us to use less energy and to develop technologies that more efficiently extract energy from a resource or that more efficiently turn that energy into the light, heat, power, etc. that we use. This increasing energy efficiency may or may not cause us to lower our fossil fuel use; we'll use less resource per unit of energy but this reduced-resource-need-per-unit-of-energy feature tends to make energy cheaper (it embodies fewer resources), so people and firms buy more energy as a result. Thus, the net effect of increased energy efficiency on resource use is ambiguous. This is the "Jevons paradox," also known as the rebound effect.

In the end, however, we can't infinitely increase our efficiency. Thus, with a growing population (though population growth is slowing) and energy demand also still growing in some parts of the world (notably, developing countries, and perhaps most notably in China and India), not all of the world's energy problems can be solved simply by conservation and efficiency gains.

Second, the increasing price will push us to switch even faster than we already are from fossil fuels to other energy sources. Other energy sources serve as a backstop technology: when energy from fossil fuels becomes too expensive, it becomes cheaper to use something else to serve our energy needs. This is why some economists don't worry much about future energy. A free market's energy prices should transmit the signal "We need more energy, and we're

willing to pay for it!” There will be money to be made in providing cheap energy, so profit-seeking firms will research and develop new energy sources. In other words, increasing prices because of increasing scarcity will inevitably cause us to switch from increasingly scarce nonrenewable (fossil fuel) energy sources to other sources as we continue to seek the lowest-cost source at every moment in time. This is what we call an energy transition.

Of course, no energy source is sufficient to meet all our energy needs. As a result, we should think in terms of an energy portfolio: our future energy needs can only be met by a mix of a variety of sources. Deliberate energy policy applies pressure to rebalance that portfolio relative to what would happen without intervention, toward a portfolio that’s more socially beneficial.

First, externalities distort this path so that with only market forces we’ll switch later and slower than would be optimal, so policy should push towards the optimal transition. Second, markets for energy and related resources are often not free markets: many of them have a lot of direct government involvement and many also have fun subsidies tucked in interesting places, and it’s hard to assess the net impact of all of these features. I still think that scarcity will inevitably exert a pressure, but that “signal” will be muted and perhaps delayed. Third, the transition from fossil fuels to other energy sources may be painful in the costs it imposes on people and firms; there may be a role for policy in easing that transition. In addition, we may want a just energy transition. Justice is implicated in the production and use of energy in a lot of ways. It is important that vulnerable and minoritized communities have access to the same improvements in environmental quality that advantaged communities get during the transition, and that they have voice, and that their access to affordable energy is not jeopardized.

So we know we will be transitioning from fossil fuels to greener energy sources increasingly over the coming years, and we know that we need research and development to improve those technologies. The government must have a role in promoting research and development in general, because innovation is a public good; and of course the externalities associated with fossil fuels make that even more sensible in the context of clean energy sources. So governments should help bring these new technologies forward. The difficulty is that we don’t know what innovations will come in the future, so we don’t know *which* energy sources will be the best in the future – so which do we subsidize? We don’t want to subsidize one energy source (say, hydrogen fuel cells) and have that turn out to be a dud. That’s what’s called “picking a winner.” We want the market to pick the winner: all other things equal, the cheapest and most efficient technology should win out because that’s what would be most profitable in a level-playing-field world. So the trick is to structure subsidies so that they reward innovation in general without preferring one technology over another. Indeed, if we simply priced the pollution and other externalities associated with the energy sources we’re trying to get off, then that would generate this same kind of incentive to innovate.

International Energy Use

We measure energy in Btu (British thermal units), which allows us to compare energy use and energy supply and whatnot across different energy sources.

Energy intensity – the amount of energy needed to produce a given amount of economic activity (btus per dollar of GDP) – has declined dramatically over time, which is good news.

This reflects energy efficiency in a broad sense: it includes doing things that take less energy (e.g., switching from manufacturing to service in the economy) and doing any given thing more efficiently.

Historically, energy use in developed (rich) countries was much greater than energy use in developing (poor) countries. There was a period where high-income countries' use seemed to peak, as the countries had relatively stable populations and mature economies and energy efficiency growth more than counteracted growth in use of energy services. Now increased demands from stuff like AI data centers is pushing back on that to some extent. But the biggest energy use growth in the world is coming in the parts of the world that are growing fastest economically. Energy use is skyrocketing in developing countries where the economies and (on average) populations are growing and they are catching up with technologies that developed countries have already adopted (e.g., expanding access to electricity and cars, building up industry, etc.). Indeed, China and India are closely watched: because of their big populations and the need for improvement in standards of living in both places, they drive lot of current and projected increases in energy use.

Within a country, energy use over time generally tracks with how strongly economies are performing; weak economies (e.g. economies in recession) produce and consume less, which generally means less energy use. When you look at energy use trends, you'll see sad little dimples at major events like the 2008 recession and the COVID-19 pandemic.

We get energy from a number of different kinds of sources. Currently, most of it comes from fossil fuel sources: liquids (including oil), coal, and natural gas.² Non-fossil fuel sources (nuclear and renewables) make up a portion of the sources, and they are increasing at tremendous rates, but they are still not our dominant energy source. Projections are for renewables to keep growing but for the fossil fuels to remain dominant for a long time. We had been expecting fossil fuels to decline steadily over time, and for a while they were declining faster than projections, but recent years have made things hard to predict. But to me there is no realistic world in which we suddenly stop burning coal and oil, and there never was.

To summarize a few important points about international energy use:

- It's growing slowly in rich (OECD) countries, but quite fast in low-income countries, and we expect these trends to continue
- We're becoming better at producing more stuff with less energy (energy intensity is decreasing)
- Fossil fuels make up a large part of our current energy portfolio, and in absolute numbers they are still growing
- Renewables are a rapidly growing share of energy sources
- Electrification of a lot of uses (e.g., transportation and heating) will probably be necessary to complete the switch to non-greenhouse gas generating energy sources

² Fossil fuels are formed by decomposition of organic matter over millions to hundreds of millions of years. They do not regenerate at an economically meaningful rate. They are dense sources of carbon, and energy locked in atomic bonds is released when the fuels are burned. Unfortunately, pollutants are also released at the same time—notably CO₂ and other greenhouse gases.

US Energy Use

The US uses a lot of energy, but energy consumption in the last decade or more has been flat, as energy needs increase at about the same rate as increases in energy efficiency. It pains me to note that for the first time in as long as I've been looking at the numbers, US energy use is projected to increase in coming years.

We import a fair amount of energy into the country, but we also export a lot – in fact, we now import and export similar amounts of oil, and we export much more natural gas than we import.

Like the rest of the world, the big sectors of our energy use are electricity, industrial, and transportation, and much smaller are residential and commercial. This is one reason I'm skeptical of climate policies that are all about tweaking home energy consumption.

Our portfolio of energy sources is similar to global averages. The shale gas revolution and the rise of renewables have caused similar trends here as in other countries. One way that we vary from some parts of the world is that in the US, we're really not building new nuclear plants; most of our nuclear energy comes from old plants that are supposed to be decommissioned.

Oil (Petroleum)

Oil is one of our favorite sources of energy, particularly for transportation; as noted above, transportation depends largely on liquid fuels, and we turn oil into gas and diesel to use it for cars and trucks and planes and boats and so on. Oil is a fossil fuel, so it is energy-dense and nonrenewable. But when we burn oil, as with other fossil fuels, we release an array of pollutants, including greenhouse gases such as CO₂ and nasty things that hang around in the local area to make smog and cause health problems.

There are many kinds and qualities of oil. Oil as it is pulled out of a well is crude; it must then be refined so it can be turned into a usable fuel source. In the US, many refineries are in the Gulf Coast region, but there are others scattered across the country, particularly in the Midwest. Refinery capacity is hard to change quickly, so whenever anything happens that affects refining or supply or demand, we can see price spikes or troughs, or even short-term shortages. How does oil get from its extraction site to a refinery? Pipelines are the cheapest and, perhaps surprisingly, probably the cleanest way – the alternatives are generally not only costly but dangerous (trucks and trains).

There are different kinds of crude oil. "Light" crude is low density and yields more fuel than does "heavy" oil; "sweet" crude has less sulfur and other nasty things in it than does "sour." Geopolitical events that affect the supply of the nicest oil, light sweet crude, have outsized impacts because light sweet crude is the best oil and the easiest to work with. As you may have noticed, lots of geopolitical events happen in places with oil.

Finally, there's a distinction between conventional and unconventional oil. Conventional oil is extracted from an oil well. This includes offshore drilling, which is costlier and probably has more externality risks than onshore wells. Unconventional oil uses other methods. One example is the "oil sands" or "tar sands" (the technical name is "bituminous sands"). These oil

sands contain heavy oil mixed with various chemicals in deposits that are very difficult (costly) to extract from. The chemicals in the oil also make it “dirtier” when the refined oil is eventually burned. There are big deposits of unconventional oil in Canada and Venezuela, among other places. Since the price of oil has risen and technology has advanced, it’s become economically viable to use these resources.

Another source of unconventional oil is “tight oil,” also known as “shale oil.” This is “fracking” (hydraulic fracturing), which you’ve probably heard of in the context of natural gas, but actually natural gas and oil are both found in underground formations of shale or other rock and can be extracted this way. I’ll discuss fracking in more detail below in the section on natural gas.

Where does oil come from? The biggest single(ish) player is the cartel OPEC (the Organization of Petroleum Exporting Countries). A cartel is a group of firms (or, in this case, countries) that agrees to manage their interaction with the market for their mutual benefit. In this case, it’s a quantity-fixing cartel: they meet regularly to decide how much each member state will produce. In this way, they limit the supply of oil, which pushes up the price and thus their profits. They say that their goal is to stabilize global oil markets, but let’s be real.

OPEC has 12 member countries.³ Saudi Arabia produces over twice as much oil as the next largest producer. Together, OPEC countries produce less than a third of the world’s oil, though they used to produce a larger share; however, they do produce a lot of the oil that’s traded globally, so they can still be influential. And of course, non-OPEC countries do not coordinate at all while OPEC tries to. However, it’s not really clear how big of an influence their cartel can have on the market; or, indeed, how well Saudi Arabia can control the rest of OPEC to try to do this quantity coordination.

Global oil production has continued to increase over recent years, mostly from non-OPEC countries.

Major non-OPEC players include the US, Russia, Canada, and China. The US now produces more oil than any other country in the world, including Saudi Arabia. Norway and the UK used to be major producers but their output is on the decline as their reserves dwindle. Overall, across OPEC and non-OPEC countries, the positions of top oil producers have varied a lot over time.

Oil production is very important to countries’ economies. In fact, in many countries, particularly in OPEC, oil production is done by a government-owned and operated entity. Many non-OPEC countries do have oil production by private companies, though. Oil production companies receive explicit and implicit subsidies in a variety of ways. (Here’s an example of an implicit subsidy: if oil is drilled from an area that the government owns, the government is entitled to some kind of “royalty” payment. Those payments are far lower than the value that firms get from the right to this extraction, so this is an implicit subsidy.)

³ Algeria, Equatorial Guinea, Gabon, Iran, Iraq, Kuwait, Libya, Nigeria, Republic of the Congo, Saudi Arabia, United Arab Emirates, and Venezuela.

Oil-producing countries often also subsidize purchase of oil by their citizens quite heavily; some argue that oil states are able to buy citizen complacency in the face of a relatively authoritarian regime through subsidies like this. Any talk about reducing the subsidization of oil is politically challenging, to say the least, not just in countries like Saudi Arabia but also here in the US.

What about oil consumption? As you might expect, US is the top consumer of oil; next is China; and no other country comes close.

Where does imported US oil come from? We used to talk a lot about suppliers we have complicated relationships with, and what that meant for our “oil security,” and that’s certainly an issue. But we generally import more oil from our neighbors Canada and Mexico than from other countries.

How is oil used? The biggest use is transportation, but a lot is used in industry. Remember, too, that in addition to consumers directly using transportation themselves, all physical items that are bought or sold also require transportation, so when oil prices rise, that feeds into the cost of consumption directly and by increasing the costs of industrial use and business transportation.

The oil market pretty important in global matters. Oil prices are very volatile, and are driven by two or three factors: supply, demand, and possibly speculation. There are short term shocks to supply, like political stuff going on where oil is produced, that can drive down supply and thus drive up the price. High demand can also drive up prices, and weak demand can drive it down; this is why oil prices usually move in tandem with the global economy.

Which is better: low oil prices or high oil prices? Well, high oil prices are good for oil sellers and bad for oil buyers. Because a lot of productive sectors use a lot of energy, oil prices that are too high can be a drag on the global economy. It’s not so bad for the US now that we produce so much oil. But another thing is that the harder-to-get-at kinds of oil, like tar sands, tight (fracked) oil, and offshore, are only economical if oil prices are high enough to cover these costs. So that’s an environmental downside of expensive oil.

Where does the price of oil come from? Oil is a globally-traded commodity, so its price is set on a global market. In some American political campaigns we hear talk about how allowing drilling in this or that place will bring gas prices back down to \$X per gallon. This is just political rhetoric. The US does produce a lot of oil so we can affect the global market price. But it is not the case that if we produce as much as we consume, we are isolated from things that happen on the global market; nor would we want it to! Most economists don’t have much interest in “energy independence” per se—the idea of producing as much energy as we use. As long as there is a global market out there, the opportunity cost of energy produced here is still the global price of energy (unless we erect massive barriers to trade, which is not a good idea).

Another thing we hear about periodically is the US president addressing a price spike by releasing oil from a strategic petroleum reserve; because the price is driven by global factors this is never going to have much of an impact, unless the spike is because of a short term localized issue like a refinery that’s gone down.

The last idea I want to discuss is “peak oil.” Since the 1970’s, people have argued that we are nearing or past the point at which the world’s ability to produce oil is peaking, and thus oil production will diminish forever after. I don’t find this a useful idea. At some point, yes, oil extraction and use will decline because it’s a nonrenewable resource. But I don’t think this a problem per se. Why? Because this is the way the allocation of a scarce resource ought to work. Producers have some idea how much oil is left in known wells and how much oil they expect to find with discovery activities, and they know that every unit extracted makes oil scarcer. This scarcity means that the opportunity cost of taking a unit of oil out right now is increasing, and thus the price of oil should rise to reflect that scarcity rent. At some point, the high price of oil will render the provision of energy from alternative sources economically profitable. At that point, we switch. The only question is whether we make that switch efficiently, and whether governments ought to be involved to ease the transition.

Now, let’s summarize some of the pros and cons of oil.

Pros:

- Cheap
- Energy-dense
- Plentiful

Cons:

- Consumption externalities:
 - Burning creates greenhouse gases and other pollutants
- Extraction externalities:
 - Drilling, particularly offshore, may result in environmental damage and costs to health and safety of workers (not externalities if the firms are held liable!)
 - Unconventional source (such as tar sands and tight oil) extraction may cause major environmental damage; the oil thus generated is also often dirtier (and thus creates worse consumption externalities when burned)
- Externalities in transporting the fuel:
 - Oil spills from pipelines, tankers, trains, or trucks (NOT external if firms are fully liable)
- Other concerns:
 - International security: oil revenues fund “problem states”
 - Energy dependence: it would be possible for major oil producers (mainly OPEC) to really hurt any particular country (even the US) if they wanted to
 - The market for oil is not perfectly competitive, although the environmental economic implications from this are not entirely bad!

Natural Gas

Natural gas is another fossil fuel, so it has some of the same characteristics as oil: it’s an energy-dense hydrocarbon. It’s a cocktail of gases, primarily methane, that are often found with deposits of other things, so it’s also called “associated gas.” Deposits of natural gas are distributed pretty widely across the globe. Some of the big uses historically for natural gas are

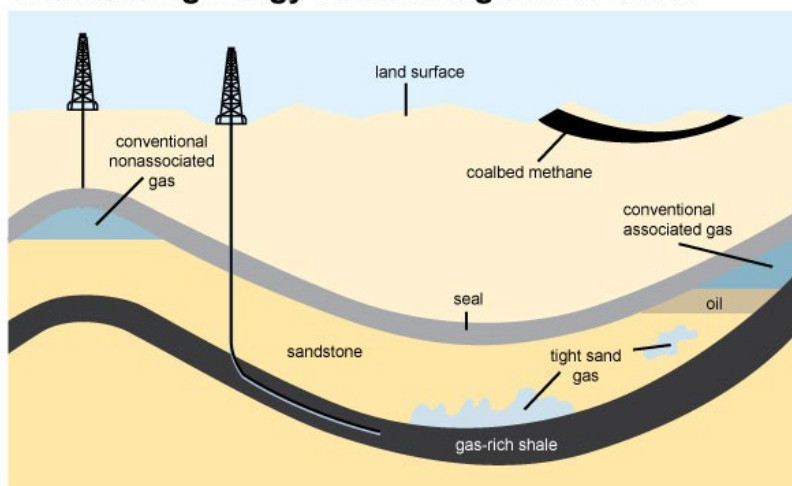
electricity and heat. It's often used near its extraction site because (since it's a gas) it's not super easy to transport. It becomes easier to transport, e.g. by pipelines, if it is converted into LNG (liquefied natural gas) or CNG (compressed natural gas).

One aspect of natural gas is that it is relatively “clean:” when burned, it produces less pollution (including less CO₂) per energy produced when compared to coal and gasoline. So not clean relative to renewables, but relative to the “worse” fossil fuels. On the other hand, if natural gas leaks at any point in the process, that's bad: leaked natural gas is a much more potent greenhouse gas than CO₂. Leaks can occur in extraction, when the gas is stored in tanks, and when the gas is transported through pipelines. It's not clear how widespread these leaks are so accounting for them is a big priority right now.

In the US, natural gas production has been transformed in recent decades to be primarily based on fracking. We don't know how extraction technology will develop over the coming decades, but at the very least we expect the role of fracked gas to stay strong, if not increase.

Remember, natural gas is often found with oil or coal. When you see fires flaring atop an oil well, you're seeing excess natural gas being burning off. That gas can be captured and sold if it is profitable to do so, but often it is not. Shale gas is found laced through shale formations like the “Marcellus Formation” that underlies parts of New York, Pennsylvania, Ohio, Maryland, West Virginia, and Virginia. Tight gas is similar to shale gas, in that both are very hard to extract, but whereas shale gas is laced into soft, fine-grained rock, tight gas is stuck amid hard, impermeable rock. In both cases, innovations in extraction have made it possible to get the gas out in cases that may have been impossible in the past. Hydraulic fracturing (also known as “fracking”) uses high-pressure water and chemicals to crack the rock in which the gas is hiding and force the gas to the surface, and this method, in combination with advances in ability to drill long horizontal shafts and discovery of vast reserves, has transformed the natural gas market.

Schematic geology of natural gas resources



Source: Adapted from United States Geological Survey factsheet 0113-01 (public domain)

Source: US Energy Information Agency website https://www.eia.gov/energyexplained/index.php?page=natural_gas_where

See the figure above to see how it works. The well on the left is a conventional well for natural gas; the one next to it is a fracked well that's targeting the “gas-rich shale” layer.

Natural gas has grown dramatically in recent years and that's predicted to continue, and that's largely because of shale gas. Natural gas is the main reason why coal has declined, and this is good for our climate goals (*if* leaks are minimal). One downside is that the energy price reductions brought about by plentiful natural gas makes us delay making investments to build up renewable energy capacity.

Hydraulic fracturing, however, is controversial. The chemicals used in the process ("fracking fluids") are proprietary and thus not publicly disclosed; some fear they are toxic, although gas companies say they are largely inert. People also worry that both natural gas and these chemicals might get into the groundwater and into wells near the drilling site. There are also concerns about disposal of the leftover water and chemicals that were used in the drilling ("produced water").

Now, let's summarize some of the pros and cons of natural gas.

Pros:

- Cheap (and may stay cheap for a while)
- Energy-dense
- Plentiful
- "Local"
- Relatively clean (for a fossil fuel)

Cons:

- Externalities from leaked natural gas at all points in the process
- Consumption externalities:
 - Burning creates pollutants including greenhouse gases (but less than other fossil fuels)
- Extraction externalities:
 - Extraction of some associated and non-associated gas involves drilling or mining, which may damage some ecosystems, and accidents can happen.
 - Extraction from shale deposits ("fracking") may be damaging because of groundwater contamination and disposal of "fracking fluids"
- Externalities in transporting the fuel:
 - Pipeline ruptures will happen occasionally (not an externality if producer is liable)
- Other concerns:
 - Geopolitics: for example, Europe relied (uneasily) on Russian natural gas pipelines for a long time, and that became a problem when the Ukraine war started.

Coal

As noted above, coal is clearly on the decline because of falling costs of natural gas and renewable energy sources. Even without regulations, we should expect this decline to continue. However, it still has an important role in global energy, so we'll consider it.

Coal is another fossil fuel, a nonrenewable energy-dense hydrocarbon. Some coal is naturally cleaner (less sulfur) and some is dirtier (more sulfur). Thus when coal is burned (generally to generate electricity), we produce not just greenhouse gases like CO₂ but also sulfur and nitrogen compounds (SO_x and NO_x). These pollutants also contribute to the regional problem of acid rain and health damages (including premature death) in local areas. When coal is burned, however, power plants can take abatement actions to reduce emissions, including installing scrubbers. “Clean coal” is the idea of coupling coal burning with a new technology called carbon capture and storage (CCS) or carbon capture, utilization, and storage (CCUS). This technology has been piloted but is really not ready to go at scale at this point, and is still very expensive.

Coal is scattered around the world in localized deposits, and is extracted, often at relatively low cost, through mines. Scarcity doesn’t seem to be a real problem with coal yet. However, extraction now happens in a variety of forms, including traditional underground mines (which can be dangerous for workers, especially in relatively low regulation settings like developing countries) and mountaintop removal (which obviously destroys a whole mountaintop, but theoretically they can somehow cover it again when they’re done).

Coal has remained a reliable source of energy worldwide in recent years. Coal use declines in the US and other OECD countries is projected to be offset by increases in China and the rest of the world, as developing countries build power plants to expand electricity access.

Now, let’s summarize some of the pros and cons of coal.

Pros:

- Cheap
- Energy-dense
- Plentiful
- “Local”

Cons:

- Consumption externalities:
 - Burning creates pollution that causes climate change, acid rain, and serious health problems
 - Clean coal technology is costly and relatively untested
- Extraction externalities:
 - Underground mines can be damaging to habitats and dangerous for workers (not external if firm is liable)
 - Mountain-top removal mining may have serious environmental externalities

Nuclear Power

Nuclear energy uses nuclear fission to break up atoms into smaller atoms, in the process releasing a lot of energy from a small amount of fuel. We use nuclear energy to generate electricity. The fuel is uranium ore that has been enriched. Uranium exists abundantly in the

earth. But since uranium can also be enriched to provide ingredients for nuclear bombs or “dirty bombs,” there’s a lot of concern about security of enriched uranium.

Nuclear power has traditionally been generated in immense reactors that perform fission on the uranium fuel. These reactors are incredibly expensive to build, so nuclear has very high fixed costs, but because each unit of fissioned fuel produces so much energy, the marginal costs of energy production are low.

There’s a lot of talk now about innovations in new smaller nuclear reactors; I always reserve judgment on new technologies until they are proven.

Nuclear energy is considered a “no-carbon” fuel source by some, since no fossil fuels are burned to generate the energy. It is considered “low-carbon” by others, since it takes energy to build and operate the plant, and that energy generally comes from fossil fuel sources. (I think that’s not quite fair, since that energy could arguably come from cleaner sources, but then again there’s a lot of concrete there, which generates carbon in its production.)

There are three main problems with nuclear energy. First, at least in its traditional huge-reactor (with massive fixed costs) form, it’s often not economical. Governments nearly always are involved in building or at least financing nuclear power plants because these high fixed costs make nuclear energy even more of a natural monopoly than other energy sources. In fact, a big part of the reason we don’t have new nuclear in the US is that it’s not worth it.

Second, folks worry about the safety of nuclear power, both from issues in their operation and from vulnerability if anything happens to the location they are sited in. Nuclear power plant designs have gotten safer over the years, but headline-grabbing disasters, with or without loss of life, over the years have energized anti-nuclear advocates. Some countries have declared an end to nuclear power, but some of those walked those plans back because of energy needs.

Third, nuclear power creates radioactive waste which will remain radioactive for hundreds or thousands of years, or much longer. Theoretically it can be stored in a very well-sealed container deep underground, but we have to consider whether these containers may leak or that unsuspecting future generations may come across them somehow. Nuclear waste that has not yet been disposed of is also a security concern: it can be stolen and used in a “dirty bomb.”

Nuclear power boomed through the 1970’s and 1980’s, and has flattened out since then; it looked like it was declining, but then picked up again in recent years.

It may be that the post-1970 increase was driven by a desire to reduce our worldwide dependence on oil (remember the 1973-4 oil embargo that was so unpleasant to the US). The decreasing rate of opening new reactors seems to have started after the 1986 Chernobyl disaster. There really have not been a lot of new nuclear plants built worldwide in the last couple of decades, though in February 2012 the US approved our first new nuclear reactors since 1978. Plants are having their operating lives extended, but many plants are finally being closed as they reach the end of the time in which they could conceivably operate.

Globally, the US produces the most nuclear energy, followed by China, France, and Russia.

Now, let's summarize some of the pros and cons of nuclear power.

Pros:

- A lot of energy from a little fuel
- Fuel is relatively abundant
- Low-carbon or no-carbon
- Very low marginal costs of energy provision once the reactors are built

Cons:

- Production externalities:
 - Small possibility of nuclear reactor accident (small chance of very large damages) – not external if firms are liable
 - Spent fuel is radioactive waste, and must be safely stored for an extremely long time (10,000 to 1 million years). Accidents can happen with this spent fuel.
- Extraction externalities:
 - Some accidents occur in mining nuclear fuel, as with any other kind of mining
- Other concerns:
 - For traditional nuclear, very high fixed (start-up) costs; government generally must be involved
 - Security concern with fuel, reactors, and waste: reactors could be terrorist targets, and the fuel and waste could be used as weapons
 - Concern about letting “trouble states” (e.g. Iran) develop nuclear power because they may enable the regime to develop nuclear weapons

Renewables

This phrase refers to a variety of sources that either can regenerate or are basically not affected by our use of them. Solar, hydroelectric, wind, and geothermal are low-carbon/no-carbon energy renewable resources: they create very little pollution in their consumption. Biofuels are renewable hydrocarbon fuels. Mostly we think about renewables with regard to electricity; they can be used for other energy needs (e.g., solar cars or biomass heat), but that's kind of at the margins; mostly, if we want to use renewables to power things, we find ways to make those things be electric-powered. And right now, solar and wind are the big ones.

Renewable energy sources used to be more expensive (per unit of energy generated) than fossil fuels, and we thought of them as the backstop technology for fossil fuels: we'd switch over from fossil fuels to renewables once fossil fuels get too expensive and/or renewable prices come down enough.

Well, this has changed a lot as the price of renewables has plummeted in response to massive technological innovation, hence their increasing adoption. In many cases renewables are cheaper than even natural gas for the marginal unit of energy produced, but this depends on a variety of factors including solar radiation, topography, etc., so it is not the case everywhere. In addition, remember that fossil fuels have an “unfair cost advantage” because of the externalities

associated with them, so we are already inclined to switch too little and too late, relative to what is optimal. In addition, some of the key renewables can only provide energy intermittently, so can't power too much of the grid without robust battery storage, which is being rolled out but is still in its early days.

Let's look at biofuels and then at the other renewables.

Biofuels

Biofuels is a broad class of hydrocarbon fuels that don't come from fossil sources. Essentially, you find a plant source, process it in some way, and then you burn it. Biofuels were long favored by some green-oriented people, but now the picture of their environmental impact is not as clear. Hence my skepticism about sustainable aviation fuel. The net carbon impact of these fuels is very tricky to calculate: for each biofuel, you grow some plants, and perhaps use fossil fuel-based fertilizers, and often use energy for tractors and whatnot. The plants absorb carbon dioxide for the time they grow; then you harvest them and burn them. And demand for biofuels may force some land that would otherwise be standing forests to be cleared periodically (for wood biofuels) or to be converted to agriculture (for corn or other agricultural biofuels). These interactions are complicated and not the sort of thing that is easily visible to the naked eye. Some analyses have shown that biofuels are not only not good, but perhaps actively bad for the environment; others debate this. Still, biofuels are renewable if they are managed sustainably, which can't be said for fossil fuels.

One popular kind of biofuel is ethanol, which is an alcohol used as a fuel or a fuel additive. It can come from corn, sugar cane, beets, or other sources. Corn is a big source of it in the US right now, and it is heavily subsidized. In some places in the US, gasoline is required by law to have a percentage of ethanol added, with the percentage varying by season, ostensibly with the goal of environmental goodness. However, it has been argued that ethanol policy in the United States is largely political: agricultural lobbies are strong, and support of farm state voters is coveted by politicians. Whenever corn prices are high, the ethanol mandates look like a particularly bad idea, feeding through into increased gas prices and taking corn out of the food market.

There's also biodiesel: vegetable oils (palm oil, corn oil, soybean oil) that can be used as a liquid fuel in diesel-type engines, including used cooking oils. My feeling is that it's nice as far as it goes, but it seems like it has to be pretty limited in scope.

Various kinds of solid biomass can also be used, including biomass from wood or waste wood (which comes largely from the paper industry). Other sources people have been exploring range from algae to switchgrass to garbage. These also seem limited to me.

Another problem with biofuels is the interaction between the production of these fuels and the production of agricultural products for food. If biofuels are grown on land that would otherwise have been used to grow food, or are derived from products that could otherwise be used for food, these biofuels will drive up the price of food. The flip side of this is that, if they are small enough to have a limited impact on food and agriculture, they can't really be producing too

much energy, so either they are driving up food prices or they don't matter much to our energy supply anyway.

Now, let's summarize some of the pros and cons of biofuels.

Pros:

- Somewhat cheap (but how cheap is hard to say because ethanol is heavily subsidized)
- Renewable
- May (or may not) be less bad for the environment on net

Cons:

- Not apparently economically competitive (i.e., they only seem to be used when there are big subsidies / mandates)
- Consumption externalities:
 - Burning creates greenhouse gases and other pollutants
- "Extraction" externalities:
 - Intensive agriculture may hurt habitats, generate carbon-type pollution, and generate water pollution
- Other concerns:
 - May drive up food prices
 - How much energy can we get from it, really?

Solar, Wind, Hydropower, and Geothermal

The other main kinds of renewable energy are solar, wind, hydropower, and geothermal. I will not discuss wave and tide power, since they don't seem to be in great use right now and I don't know much about them. All of these sources are "no-carbon" or "low-carbon" in just the same way that nuclear power is: in operation they do not generate greenhouse gases, but to start generating energy, the plants (or panels or windmills) must be built, which takes energy, which often comes from fossil fuels. Again, as our energy portfolio changes over the next few decades, this will be much less true. And the fixed infrastructure for solar and wind is relatively low.

Hydropower is an old source and is the most established, and most used, of the renewable energy sources. It captures kinetic energy from flowing water and turns it into electricity. This often involves damming up a water source, and thus changes the flow of a river. These changes, which often include flooding one area and limiting the flow where the river has historically run, can be quite environmentally damaging and can have tremendous effects on people living either in the area of the dam or in downstream areas where flow patterns are changed. The good thing about hydropower is that of the renewables, it's one of the few that's really reliable for generating a lot of power all day – that is, it's not intermittent. On the other hand, hydropower plants also are capital-intensive to build (high fixed cost and low marginal cost), and thus governments tend to have to be involved. Obviously, hydropower only works in some areas; the electricity thus generated must be transported to regions that don't have access. Hydropower plants will need to consider changing patterns of water availability caused by climate change. I've heard some talk about small dams or even microdams, but that conversation doesn't seem to have gotten too far.

Solar energy captures energy from the sun and makes it available to us, usually in the form of electricity or sometimes heat. Electricity from solar energy generally uses panels of photovoltaic cells to convert energy from the sun into electricity. Solar energy has improved over the years, and in some places is cost-competitive. This is in large part because China subsidized and promoted solar industry innovation. This also put competitive pressure on solar producers in other countries. There is some really nasty pollution associated with the construction of solar panels, so controls must be in place for that. While solar panels have become more efficient every year, you still need to cover a lot of area with panels to get a meaningful amount of energy. Should solar panels be distributed across rooftops? Would that be enough? Probably not. Should we build massive solar farms in the desert? That could have a big environmental impact on habitats and species. But solar has been expanding faster than anyone projected and is expected to continue to expand for the foreseeable future.

The other big problem with solar energy is that it only works where and when the sun shines, i.e., it is intermittent. We must collect the energy while the sun shines and then store it so we can use it later. Grid scale and house scale battery storage is being rolled out, but there's not nearly enough and it's not efficient enough, so more work is still needed on battery storage.

Wind power uses wind turbines to capture kinetic energy from wind and convert into electricity. There are few externalities associated with it, although bird populations can be hurt (less with newer turbines, which spin more slowly, than with the older ones). While residential wind is not really as much of a thing as residential solar, starting up a wind farm doesn't require as big of an investment up front as some energy sources because wind farms can be built in a piecemeal fashion (you can add a windmill at a time). Some people feel that windmills destroy a landscape's visual or spiritual qualities; others say that the sound of massive windmills is disturbing and some say causes ill health effects (it's called "wind turbine syndrome," and evokes a lot of skepticism); so wind installations get a lot of protest. Here in Massachusetts, major wind projects offshore near Boston (Cape Wind) and proposed ones out here in the Berkshires have been very controversial. Wind power still is also becoming cost-competitive. Wind is also intermittent and also only works in some places, and so storage technology must improve. In addition, I understand that windmills still tend to break down a lot. Offshore wind has more continuous and stronger wind, but then the maintenance is even harder. Anyway, like solar, wind has grown a lot in recent years, both in the US and internationally.

Geothermal energy converts the energy of subterranean heat into electricity. One way of getting this energy out is to find areas where geothermal steam is naturally available at the ground level, and another is to generate geothermal steam by pumping water into the ground through one hole and capture the heated steam that comes out another hole. You then essentially extract energy from the steam. It's not in terribly great use worldwide, although some countries use it quite a bit (Iceland gets 65% (!) of their energy from geothermal,⁴ and the Philippines uses a lot as well). Geothermal has high fixed costs to build a plant but may have low marginal costs of generation. It's not clear to me how much potential geothermal has as a worldwide energy source—it depends a lot on the geological characteristics of a region.

⁴ <https://www.government.is/topics/business-and-industry/energy/geothermal/>